$$E_{\text{ph}} + E_0 = E_1 + \Delta U,$$

or

$$E_{\text{ph}} = (E_1 - E_0) + \Delta U. \tag{6.11}$$

It must be that the photon does *not* arrive at the floor with just the energy $E_1 - E_0$ it started with, but with a *little more energy*. Otherwise some energy would have been lost. If we substitute in Eq. (6.11) the $\Delta U$ we got in Eq. (6.10), we get that the photon arrives at the floor with the energy

$$E_{\text{ph}} = (E_1 - E_0)\left(1 + \frac{gH}{c^2}\right). \tag{6.12}$$

But a photon of energy $E_{\text{ph}}$ has the frequency $\omega = E_{\text{ph}}/\hbar$. Calling the frequency of the *emitted* photon $\omega_0$—which is by Eq. (6.7) equal to $(E_1 - E_0)/\hbar$—our result in Eq. (6.12) gives again the relation of (6.5) between the frequency of the photon when it is absorbed on the floor and the frequency with which it was emitted.

The same result can be obtained in still another way. A photon of frequency $\omega_0$ has the energy $E_0 = \hbar\omega_0$. Since the energy $E_0$ has the gravitational mass $E_0/c^2$ the photon has a mass (*not* rest mass) $\hbar\omega_0/c^2$, and is "attracted" by the earth. In falling the distance $H$ it will gain an additional energy $(\hbar\omega_0/c^2)gH$, so it arrives with the energy

$$E = \hbar\omega_0\left(1 + \frac{gH}{c^2}\right).$$

But its frequency after the fall is $E/\hbar$, giving again the result in Eq. (6.5). Our ideas about relativity, quantum physics, and energy conservation all fit together only if Einstein's predictions about clocks in a gravitational field are right. The frequency changes we are talking about are normally very small. For instance, for an altitude difference of 20 meters at the earth's surface the frequency difference is only about two parts in $10^{15}$. However, just such a change has recently been found experimentally using the Mössbauer effect.* Einstein was perfectly correct.

* R. V. Pound and G. A. Rebka, Jr., *Physical Review Letters,* vol. 4, p. 337 (1960).